



UNIVERSITY OF ENGINEERING AND TECHNOLOGY, TAXILA

Department of Computer Engineering

FINAL YEAR PROJECT

PROGRESS REPORT

WEEK 01

Name:	Ali Sher Haidari Ahmad Haseeb Zohaib Ahsan
Roll NO:	23-CP-83 23-CP-42 23-CP-39
Internship Domain:	Conscious Robotic Exploration Using Aerial Map Priors

1. Executive Summary

During Week 1, the core infrastructure for the *Conscious Robotic Exploration* project was established. All three team members completed their Git/GitHub SSH configurations, initialized their respective track module folders according to the project workspace architecture, set up local Python/ROS environments, performed track-specific literature reviews, and submitted preliminary prototypes for their assigned tasks.

2. Team Member Accomplishments & Progress

Member 1: Ali Sher Haidari (Reg No: 23-CP-83)

- **Track: Track 1 — Aerial Vision & Cost-Map**
- **Module Folder: costmap/**
- **Key Accomplishments:**
 - Git & GitHub Foundations: Set up SSH key authentication and gained access to the central repository.
 - Module Directory Initialization: Initialized the costmap/ directory structure containing data/, models/, and scripts/ (preprocess.py, train_segmentation.py, generate_costmap.py).
 - Environment Setup: Configured local Python environment with PyTorch, TensorFlow, and OpenCV for perception tasks.
 - Literature Review: Reviewed 2 core papers focusing on aerial-to-cost-map generation and terrain semantic segmentation techniques.
 - Dataset Research: Shortlisted 2–3 UAV and satellite imagery datasets, evaluating them for licensing, resolution, and terrain class labels.

Member 2: Zohaib Ahsan (Reg No: 23-CP-39)

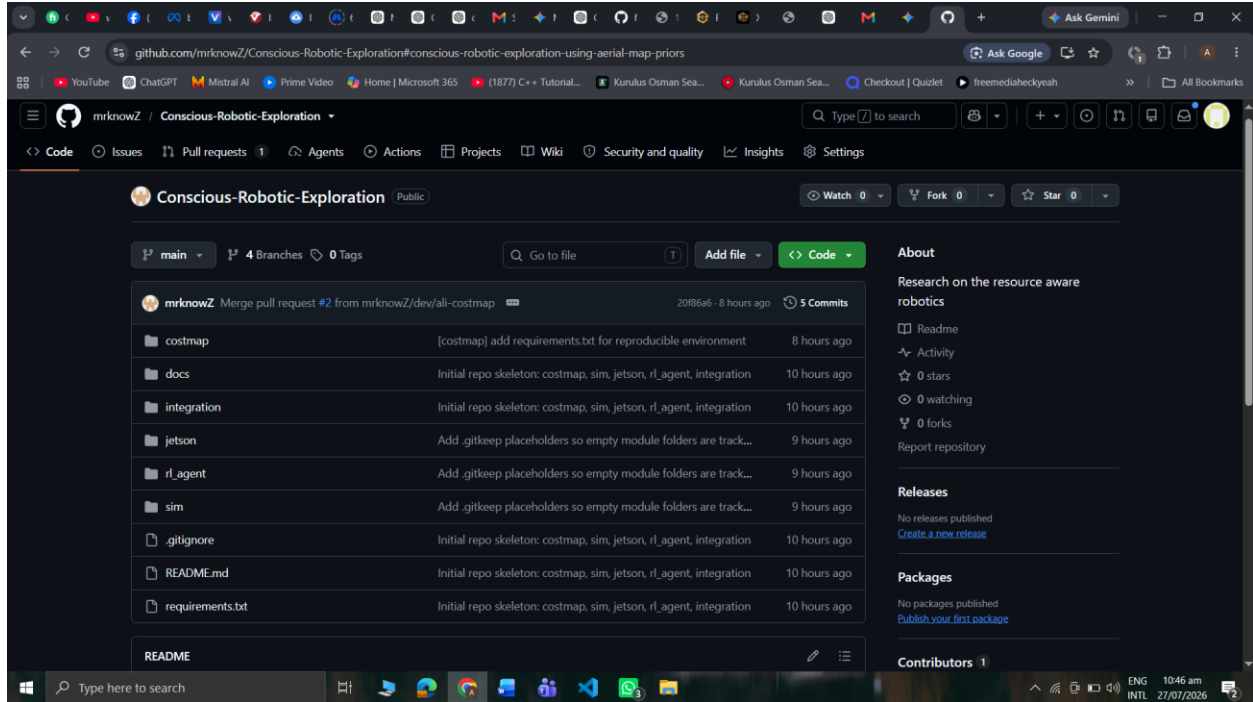
- **Track: Track 2 — Simulation, Robot & Jetson Integration**
- **Module Folders: sim/ & jetson/**
- **Key Accomplishments:**
 - Git & GitHub Foundations: Set up SSH key authentication and joined the shared repository environment.
 - Module Directories Initialization: Created the sim/ module (airsim_configs/, ros2_bridge/) and jetson/ module (deployment_scripts/, benchmarks/).

- Environment Preparation: Set up Python workspace and prepared dependencies for upcoming ROS 2 installation.
- Literature Review: Reviewed 2 research papers covering ROS 2 / AirSim bridge integration and embedded edge deployment.
- Hardware & Sim Requirements: Researched system hardware requirements for hosting Unreal Engine + AirSim simulation pipeline.

Member 3: Ahmed Haseeb (Reg No: 23-CP-42)

- **Track: Track 3 — Reinforcement Learning & Energy Model**
- **Module Folder: rl_agent/**
- **Key Accomplishments:**
 - Git & GitHub Foundations: Set up SSH keys, created local branch dev/ahmed-rl, and initialized git tracking.
 - Module Directory Initialization: Created the rl_agent/ directory structure containing envs/, energy_model/, and training/.
 - Deep Learning & RL Environment Setup: Configured virtual environment with PyTorch, Gymnasium, and Stable-Baselines3.
 - Literature Review: Conducted literature review on 2 papers regarding energy-aware RL navigation and reward shaping mechanisms.
 - Energy Model Draft & RL Prototype: Designed an initial sketch for the battery energy depletion model inside rl_agent/energy_model/energy_model.py and implemented a 5×5 toy gridworld custom Gymnasium environment in rl_agent/envs/environment.py.

3. Project Workspace Set Up



4. Installed Dependencies & Technology Stack

- Perception & Vision (Track 1): PyTorch, TensorFlow, OpenCV, NumPy
- Simulation & Embedded Systems (Track 2): ROS 2 dependencies, AirSim setup scripts, Jetson deployment tools
- Reinforcement Learning (Track 3): Gymnasium, PyTorch, Stable-Baselines3
- Version Control & Docs: Git, SSH, GitHub Pull Requests, Markdown documentation

5. Pull Request & Merging Workflow

- Governance Rule: Merging into main requires approval from at least 2 team reviewers (Review required).